

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Designed and scaled a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients**, processing **10,000+ monthly submissions** via REST integrations; contributed to **\$340K+** in revenue.
- Built a full-stack platform (**React**, Node.js, MongoDB) used daily by the AI team to prepare ML training data, cutting manual work by **100+ hours/month** and lifting throughput by **60%**.
- Hardened the backend with AWS KMS encryption, auth, and rate limiting, and remediated **20+ VAPT findings**, and piloted a COO-sponsored GitHub Copilot experiment to ship faster.
- Owned the eTriage backend as primary technical POC in a fast-paced team, debugging production incidents to root cause and shipping fixes and features end to end.

Cario Growth Services, Backend Developer Intern | *Chennai, India* Nov 2022 – May 2023

- Designed and shipped **20+ RESTful APIs** (Node.js, Fastify, PostgreSQL) with validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated core repositories from JavaScript to TypeScript and shipped a production feature running a **4-bit quantized Llama-7B** model via llama.cpp with async queued generation.

Freelance / Open Source, Full-Stack Engineer | *Remote* Jul 2022 – Aug 2023

- Built and deployed **Underdogs Fitness** (Website), a production MERN (**React** front end) product with REST APIs and cron automation, shipped independently end to end; live **3+ years**.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a full-stack agentic web-discovery product in **Python** turning open-ended prompts into structured entity tables via typed retrieval planning, reranking, and deterministic extraction with LLM fallback; hardened with **150+ tests** and shipped as a live demo.

cf-ai-research-scout – Edge-Native AI Assistant (GitHub) TypeScript, Cloudflare Workers AI, Durable Objects, D1

- Built a full-stack **TypeScript** AI assistant with durable, multi-step workflows that survive process restarts and tool failures, streaming results to the client over WebSocket in real time.

Sonare – Offline Sign / Speech App (GitHub) React, Electron, FastAPI, MediaPipe, whisper.cpp

- Engineered a **React** front end and FastAPI back end integrating camera, microphone, and local ML services (MediaPipe hand tracking, whisper.cpp ASR, local TTS) into one product, with a responsive component-based UI kept fluid under live streams.

Publications

- “Alzheimer’s Disease Classification using Transfer Learning,” IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages: Python, TypeScript, JavaScript, C++, Java, SQL, Bash

Frontend & Full-Stack: React, Next.js, Node.js, Express, FastAPI, REST APIs

ML & AI: PyTorch, Hugging Face Transformers, sentence-transformers, LLM agents, RAG, evaluation

Systems & DevOps: Data structures & algorithms, Docker, Kubernetes, distributed systems, Git, GitHub Actions, CI/CD

Cloud / Platforms: AWS, GCP, Azure, Linux, Cloudflare Workers, MongoDB, PostgreSQL

Achievements

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.